SEC P vs NP log 2026-04-25

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-25 01:12 UTC

SEC P vs NP — daily log 2026-04-25

4 cycles recorded

Finite Geometry Line Count Bounds EF Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite geometry (projective planes over GF(2)) × Extended Frege proof size
- **Recorded**: 2026-04-25 00:09 UTC
- **Entry ID**: `10350f5e2ac2`

Statement

For any tautology φ in 3-CNF, the number of lines in the projective plane PG(2,2^k) constructed from φ's clauses is Θ(2^{n/2}) if and only if the EF proof size of φ is Ω(n^2).

Rationale

Finite geometry provides a combinatorial framework to encode clause interactions, while EF proof size measures logical complexity. The projective plane's line count captures structural constraints that may correlate with proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 14, 'conjecture_hol
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 11, 'conjecture_hol
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 5, 'conjecture_hold
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 8, 'conjecture_hold
TRIAL: {'metric_name': 'EF proof size', 'metric_value': True, 'instances_tested': 11, 'conjecture_hol

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d4bc630.py", line 164, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d4bc630.py", line 164, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
```

```
                                          ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

The test failed to find a counterexample (empty 'counterexample' field) and the small instance sizes ($n \leq 14$) cannot validate asymptotic claims about $\Omega(n^2)$ vs $\Theta(2^{n/2})$ growth rates. | next: Test with larger n values (e.g., $n \geq 20$) and verify asymptotic behavior using rigorous complexity analysis

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Dimension of Variety from Tautology Ideal Bounds EF Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry (Krull dimension of polynomial ideals) × Extended Frege proof size
- **Recorded**: 2026-04-25 00:39 UTC
- **Entry ID**: `aba72d8e95aa`

Statement

The Krull dimension of the variety defined by the ideal generated by the clauses of a tautology is Θ(log size of Extended Frege proof for that tautology).

Rationale

The Krull dimension reflects the intrinsic complexity of the system's solution space, which may impose structural constraints on the proof complexity in Extended Frege.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bb1d86ad.py", line 91, in <module>
    seeds = [int(arg) for arg in sys.argv[1:]] or [11, 23, 37, 53, 71]
                                 ^^^
NameError: name 'sys' is not defined. Did you forget to import 'sys'?
```

Judge reasoning

The test code failed to execute due to a missing 'sys' import, yielding no empirical results. Without execution, we cannot assess support or falsification. | next: Fix the test code by importing 'sys' and re-run the experiment with proper setup.

[RETROACTIVE INVALIDATION] Frobenius-Schur Indicator of Clause-Symmetry Group Predicts SAT Symmetry Breaking Cost

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation theory × CDCL symmetry-breaking

- **Recorded**: 2026-04-25 01:12 UTC
- **Entry ID**: `e006a48b37a7_correction`

Statement

This entry_id originally marked SUPPORTED but audit revealed: (1) seeds ignored - identical output per seed, (2) only 3 hand-picked symmetry groups tested, (3) ratio 0.626 is tautological for cyclic_shift construction where tau=n and sb=n-1 by design.

Rationale

Manual audit by Claude and Ludo on 2026-04-25

Novelty

- Judge: `?` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Retroactive invalidation: zero multi-seed trials, deterministic per-construction output, metric was combinatorial tautology.

[RETROACTIVE INVALIDATION] Ideal Generators Count Bounds Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Commutative algebra (GF(2) polynomial rings) × Communication complexity
- **Recorded**: 2026-04-25 01:12 UTC
- **Entry ID**: `b43a4129e5c5_correction`

Statement

This entry_id originally marked SUPPORTED but audit revealed: test stdout was only the literal placeholder from the prompt template, indicating the LLM wrote a stub that printed the sample line without running any real test.

Rationale

Manual audit by Claude and Ludo on 2026-04-25

Novelty

- Judge: `?` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Retroactive invalidation: zero real test execution, stdout was just the prompt placeholder.